Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
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Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My website will be about a prosthetic limb recommendation tool. The goal I am trying to address is to create a website that helps someone explore different prosthetic options. I want my website to be able to meet their needs and let them figure out which prosthetic lets them keep doing normal activities that they love.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are a couple of similar websites that try to address this same topic, such as PROTEOR Prosthetic Foot Finder, My Prosthetic Priorities, and Ottobock Prosthetic Foot Filter. They all address the problem in different ways and focus on more on the person rather than the actual prosthetic.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

The purpose of my website is to help individuals explore prosthetic limb options based on

their lifestyle, physical activities, and personal goals. Users would answer questions about their activity level, preferred sports, and desired features, and the website would suggest prosthetic options that may fit their interests. The goal is to educate users about different prosthetic designs and help them identify options to discuss with a prosthetist, rather than provide medical advice or make clinical decisions.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The program will take user input about their activity level, preferred sports, and prosthetic features, process that information by filtering a list of prosthetic options using conditional statements and loops, and display recommendations that match the user's lifestyle and personal goals. The program will collect the user's preferred sport, activity level, and desired prosthetic features, and compare these inputs against a list of prosthetic options

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Preferred sport or activity: Running, football, hockey, walking, or hiking.
Activity level: Low, moderate, or high.
Type of prosthetic limb: Arm or leg.
Desired features: Stability, flexibility, comfort, or durability.
Primary goal: Everyday mobility, athletic performance, or recreational activities.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program will compare the user's inputs against a list of prosthetic options. It will use loops and conditional statements (if statements) to filter out options that do not match the user's selected limb type, activity level, sport, and desired features. It will create a list of prosthetic options that match the user's preferences to display as recommendations.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will receive a list of recommended prosthetic options that match their selected sport or activity, activity level, limb type, and desired features. If no options match the user's preferences, the program will display a message explaining that no matching options were found. Each recommendation will include the prosthetic's name, a brief description, and its intended uses.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

- **User Preferences:** The user's selected sport or activity, activity level, limb type, and desired prosthetic features. - **Prosthetic Limb:** Each prosthetic option's name, type, intended activity, and features. - **Recommendation:** The list of prosthetic options that match the user's preferences.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need: Name of the prosthetic Type of limb (arm or leg) Intended activity or sport Features (stability, flexibility, comfort, or durability) **Details I don't need:** Color and overall appearance Manufacturer's location Date the prosthetic was created Marketing descriptions or advertisements

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Creating the recommendation logic, like developing conditional statements that accurately match the user's activity level, sport, limb type, and desired features to suitable prosthetic options. Also, organizing the data and creating dataclasses and a list to store prosthetic information in a way that makes it easy to search and filter. Possibly another difficulty could be handling multiple preferences, like ensuring the program can evaluate several user preferences at once and return only prosthetics that match the selected criteria.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will ask the LLM to explain code in simple terms and break it into smaller steps rather than copying an entire solution. I will make sure I understand how the dataclasses, lists, loops, and conditional statements work before using them. I will compare the generated code with my project requirements and course materials to check for errors or incorrect information. I will write comments in my own words to explain the purpose of important sections of the code. (aka I will document on my own)

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My ultimate goal is to improve my understanding of Drafter by building a functional, interactive prosthetic recommendation website. I also want to add my own creative touch by designing a user-friendly experience focused on sports and biomechanics. I hope that I can also learn more about a topic I am interested in. I really like learning about prosthetics, so I think this will help grow my interests.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I asked a couple of questions: If you were looking for a prosthetic limb, what information would you want to know before choosing one? What would you want to see on a page displaying your recommendations? What might confuse you when using this website?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

They suggested that they would want clear information about each prosthetic's features, intended activities, and comfort. They also said the recommendations page should include pictures, descriptions, and explanations of why each option matches my preferences. To avoid confusion, the website should use simple language, clear questions, and easy-to-follow instructions. When I was talking to AI they mentioned some of the websites I found which were PROTEOR Prosthetic Foot Finder. My friend that I asked did not really know of an existing solution because she has little knowledge of prosthetics.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

The most complicated feature would be providing personalized recommendations based on medical research and predicting which prosthetic would work best for each user. This would be too advanced for an introductory programming project because it requires specialized medical knowledge and complex prediction methods. A simpler alternative would be to include pictures and detailed comparisons of prosthetic options so users can explore their features and intended activities.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

My project idea has become more focused on helping users explore prosthetic limbs based on their preferred sports, activity levels, and desired features. I plan to use lists, loops, and conditional statements to generate recommendations rather than complicated AI or medical predictions.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck, I will go to TA office hours and my teacher's office hours. I can also turn to friends in the course cause I might just need a new pair of eyes looking at my code.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

1. Read the error message carefully to identify what might be causing the problem.
2. Test one change at a time to see if it fixes the problem.
3. Ask the LLM for guidance by sharing the error message and asking for a simple explanation instead of copying code I do not understand.
4. Retest website
5. Ask my instructor or consult course materials if I cannot resolve the issue on my own.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

can you help me come up for a code idea that would statify these requirements. do not write any code yet though just ideas. I am interested in football, biomechanics (prosetics esspecially), and ice hockey. So those are some things my code could be about
Yes. Based on the requirements in your screenshot, I'd choose an idea where the dataclasses, list, loop, inputs, and if-statements happen naturally, rather than trying to force them into the project afterward.

Here are some ideas connected to football, biomechanics/prosthetics, and hockey. I'll keep this completely at the idea/planning level—no code yet.

1. 🦿 Prosthetic Limb Recommendation Tool — my top choice
You could create a website that helps someone explore different prosthetic options based on their needs and activities.

For example, a user could enter information such as:

Activity/sport: running, football, hockey, everyday use, etc.
Desired activity level
Prosthetic type or feature they're interested in
Maybe budget or priority, such as comfort vs. performance
Your site could then go through a collection of prosthetic options and recommend ones that fit the user's selections.

Possible dataclasses:

State could contain things like:

user's selected activity
user's activity level
user's priority
list of prosthetics
current recommendation
Then your second dataclass could be a Prosthetic, with information such as its name, intended activity, flexibility/stiffness level, and description.

This one fits the requirements particularly well because you could naturally have 7+ routes, such as Home → About Prosthetics → Questionnaire → Activity Selection → Priorities → Recommendations → Prosthetic Details → Results/Summary.

Your if statements could determine recommendations, and your loop could search through the list of prosthetics.

Difficulty: ★★★☆☆

2. 🏈 Football Player Performance / Combine Analyzer
This could be a website where someone enters a football player's athletic measurements

and gets an assessment.

The user might enter:

Position
40-yard dash time
Vertical jump
Bench press reps
Height/weight
The website could give feedback such as whether certain measurements are strong for the player's position and identify strengths.

Your second dataclass could be Player, containing information about each athlete. Your State could store a list of Player objects, allowing the user to add multiple players and compare them.

That gives you a very natural loop: go through all the players to find things like the fastest player, highest vertical, or best performer.

Possible pages could include Home, Add Player, Player List, Player Details, Compare Players, Results, and About the Measurements.

Difficulty: ★★☆☆☆ to ★★★☆☆

This would probably be the easiest sports idea to program while still having a legitimate purpose.

3. 🏒 Hockey Team / Player Performance Tracker
You could make an application for a coach or fan to keep track of hockey players.

A user could add players with things like:

Name
Position
Goals
Assists
Shots
Games played
The site could then identify things such as the team's leading scorer or highest-performing player.

Your second dataclass could naturally be HockeyPlayer, while State contains a list of those players.

You'd have lots of easy opportunities for meaningful conditions:

Is the player a goalie or skater?
Has the player scored above a certain amount?
Does the team currently have any players entered?

And looping through the roster would easily satisfy the loop requirement.

Difficulty: ★★☆☆☆

4. 🧠 Football Concussion / Helmet Education Tool
Since you're interested in biomechanics, you could combine that with football.

Instead of trying to actually diagnose concussions, which would make the project unnecessarily complicated, your site could be an educational application about helmet technology and impact protection.

A user could explore different fictional/example helmet designs based on things like:

Helmet type
Padding system
Weight
Intended position
Impact-protection category
Your application could let users compare helmet designs and select what characteristics matter to them.

Your second dataclass could be Helmet, and State could contain a list of helmets.

You could have pages such as Home → How Helmets Work → Add/Select Helmet → Helmet List → Compare → Recommendation → About Concussions.

Difficulty: ★★★☆☆

This could also connect nicely to a research topic if you've previously looked at newer football helmet technology.

5. 🦿🏈 Adaptive Athlete Equipment Finder
This is probably the most unique idea.

Your website could help an adaptive athlete explore equipment for different sports. Someone might select:

Sport: football / running / hockey
Equipment need: prosthetic leg / prosthetic arm / adaptive equipment
Goal: speed / stability / comfort / mobility

Your application would then display appropriate equipment options from a list.

You could have an Equipment dataclass with fields such as name, sport, type, purpose, and description. State could hold a list of Equipment objects plus the user's selections.

That would make your loop useful because you could iterate through all the equipment and find the options matching the person's selections.

Difficulty: ★★★★☆

It's a little more ambitious, but it combines all three areas you're interested in.

How I'd narrow it down
I think your three strongest options are:

Idea	Coding difficulty	Easy to meet requirements?	Uniqueness
🦾 Prosthetic Recommendation Tool	Medium	⭐⭐⭐⭐⭐	⭐⭐⭐⭐⭐
🏈 Football Performance Analyzer	Easy–Medium	⭐⭐⭐⭐⭐	⭐⭐⭐
🏒 Hockey Performance Tracker	Easy	⭐⭐⭐⭐⭐	⭐⭐⭐

For your particular assignment, I'd lean toward the Prosthetic Recommendation Tool. It gives you a legitimate reason to have multiple pages, inputs, conditions, a second dataclass, a list of objects, and a loop. Most importantly, those requirements wouldn't feel artificially added just to satisfy the rubric.

A manageable version could have a simple goal like:

A website that allows users to explore different prosthetic designs and recommends options based on the activities and features they select.

That is big enough to meet the project requirements without requiring an extremely complicated program.

If you like that idea, we can next brainstorm what the 7 routes, 5 pages, 2 dataclasses, 3 inputs, 3 if-statements, and 1 loop could each do, still without writing any code.

I chose the prosthetic limb recommendation tool could you now help me search the interenet for existing tools or websites currently attempt to address this same goal
Yes. I searched specifically for tools that do something close to your idea: asking about a person's needs, lifestyle, activity level, or physical characteristics and then helping narrow down an appropriate prosthetic option. There are several, but they approach the problem differently—which is useful because you can explain how your project is inspired by existing solutions without simply copying one.

1. PROTEOR Prosthetic Foot Finder — probably the closest match
PROTEOR Prosthetic Foot Finder

This is probably the best existing website for you to study. It has users take a quiz and then

recommends prosthetic feet based on their answers. PROTEOR says the goal is to account for people's different lifestyles and goals, while also making clear that the recommendation is only a general guide and a healthcare professional should make the actual clinical decision.

Connection to your project: This is extremely similar to the basic structure we discussed:

User answers questions → program evaluates answers → matching prosthetic options are shown.

For your class project, you could make a simplified educational version rather than trying to make genuine medical recommendations.
2. My Prosthetic Priorities — focuses more on the PERSON
My Prosthetic Priorities

I think this one is particularly useful for developing your project idea.

Instead of simply saying "here's the prosthetic you should get," My Prosthetic Priorities helps people think about what matters to them before choosing their first lower-limb prosthesis. Its Decision-Making Aid considers things such as lifestyle, goals, priorities, and questions the person should discuss with their prosthetist. MyProstheticPriorities

That suggests your website could ask more interesting questions than just physical measurements.

For instance, you could have users think about daily activity, sports, terrain, comfort, stability, appearance, and their main goal for using the prosthesis.

That would make your project feel more like a real decision-support tool rather than just a product catalog.
what is a good questions to ask a friend to help me brainstorm.
Here is a question to ask a friend to help you brainstorm your prosthetic limb recommendation website. I'd focus on something that helps you discover useful features and make your website stand out.
If you were an athlete looking for a prosthetic limb that would help you participate in your favorite sport, what information and features would you want my website to provide to help you explore your options?
This question is especially useful because it connects your interests in sports, biomechanics, and prosthetics while helping you identify features you could realistically implement in Drafter.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.